Amine AKHLAFA

Seeking a 6-month end-of-studies internship in Offensive Security, from March 15th, 2027

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Education

École des Mines – One of France's leading engineering universities. Nancy, France

Major in Computer Science ('Formation Ingénieur Civil des Mines') *Sept 2024 – Present*

- **Relevant coursework:** Algorithms and Complexity, Foundations of Computing, Computer Architecture, Software Engineering, Introduction to C/C++.

University of Lorraine – Dual Master's Degree in Cyber Security Nancy, France

MSc in Cyber Security (Master SIRAV - IT Security, Networks and Virtual Architectures) *Sept 2026 – Present*

- **Relevant coursework:** Network and Web Protocol security, Malware analysis, System monitoring, Access control, Infrastructure and Virtualization.

Experience

Research Project: Automating Penetration Testing with AI Agents Nancy, France

Supervised by Dr Jean-Yves Marion, LORIA-ENSMN, GODZILLA - France 2030 *Oct 2025 – June 2026*

- Designed and fine-tuned a dense retrieval pipeline (Qwen3-Embedding-0.6B, contrastive loss with per-epoch hard-negative mining) mapping offensive action descriptions to **MITRE ATT&CK** techniques, achieving **89% Hit@3** accuracy across 697 candidate TTPs.
- Open-sourced the fine-tuned model and training/evaluation datasets on **Hugging Face**, and contributed the processing pipeline to a broader agentic pentesting research effort.

Cybermines - The school's cybersecurity club (11 members) Nancy, France

President *July 2025 – June 2026*

- Designed and led a weekly offensive security curriculum covering the cyber kill chain and web application attacks; ran beginner CTF sessions and mentor members on TryHackMe labs.
- Mentored newcomers with no prior security background, breaking down foundational concepts and guiding them through their first CTF challenges to build confidence and independence.

Personal Projects

Penetration Testing Lab Writeups on GitHub

- Currently completing the **HTB Academy Penetration Tester Job Role Path**, covering web application attacks, Active Directory compromise (enumeration, Kerberos abuse, ACL attacks), password attacks, tunneling/pivoting, and file transfer techniques, documenting each stage as reusable cheatsheets and methodology notes in a personal Obsidian knowledge base.
- Built and operate a **multi-VM isolated lab** (Kali attacker, DVWA/Juice Shop and Windows 7 x86 targets) with host-only networking and static IPs, simulating real-world engagement network segmentation.
- Designed and Dockerized two **vulnerable mini-labs** (LFI via FTP file upload; SMB password spraying via exposed Git credentials) to study misconfiguration causes, practice and document full attack chains.

Skills

Programming & Systems: Python, SQL, Bash, LaTeX, Git, VMware, Obsidian.

Offensive Security Tools: Nmap, Burp Suite, Metasploit, ffuf, Netcat, NetExec, Hashcat, BloodHound.

Concepts: MITRE ATT&CK, Active Directory attacks (Kerberoasting, DCSync), web application exploitation (SQLi, XSS, LFI), post-exploitation, lateral movement (Pass-The-Hash, Pass-The-Ticket), RAG pipelines, and LLM integration.

Certifications & Training

TryHackMe (Top 1%): Jr Penetration Tester, Offensive Pentesting, CyberSecurity 101.

OSCP candidate: exam scheduled for December 2026.

Project Management: MOOC Gestion de Projet.

Languages

French (native), Arabic (fluent), English (C1 - 7.5 IELTS), Chinese (HSK2).